

AI335L Deep Learning Lab

Experiment No. 12

Deadline: June 07, 2026 12:00 AM

Lab Task (Phase 5)

Rigorous Evaluation: Ablations, Errors, and Limitations

1. From Building to Understanding

The previous four weeks were about building. This week is about understanding what you built. A trained model with a single accuracy number is a starting point, not a result. By Friday, you should be able to defend not only what your model achieves but how it fails, which of its components actually contribute, how it compares to alternatives, and where it should not be trusted.

This is the science week. The bar shifts from “does it train” to “is it real”. Two things make this week pay off: rigorous experimental design and ruthless honesty about limitations. A team that reports 81% with confidence intervals, three ablations, and a documented failure mode scores higher than a team that reports 89% with no analysis.

2. The Test-Set Ceremony

You have not touched your test set since Phase 2. This week, you touch it exactly once. After that, you do not touch it again. Whatever number comes out is your reported number.

Procedure:

- On Monday or Tuesday, freeze your final model configuration. This is the model from Phase 4 with all its hyperparameters, weights, preprocessing, and post-processing settled.
- Commit this configuration to git with a clear message: “freeze: final config for Phase 5 test evaluation”
- Run the test set evaluation. Record metrics with at least three random seeds to capture variance from any non-determinism (data shuffling, dropout at inference, etc.)
- Tag this commit: `git tag final-test-evaluation && git push --tags`
- From this point on, the test set is closed. Any further changes to the model do not get re-evaluated on test. They live on validation only and the report explicitly says so.

Why this matters: repeatedly tuning against the test set is how researchers fool themselves. The number you eventually report becomes meaningless because it implicitly absorbed information from many “final” evaluations. We enforce the ceremony so your reported number actually means something.

If the test number is disappointing: you report it honestly. Section 8 of the report (“Limitations”) is where this gets discussed. A 4-point drop from validation to test, openly diagnosed, scores higher than a hidden one.

3. Learning Objectives

By the end of Phase 5, you will be able to:

- Conduct a single, disciplined final evaluation on a held-out test set
- Design and run ablation studies that isolate the contribution of each component
- Perform structured error analysis that maps failures to causes
- Apply statistical reasoning to results: confidence intervals, significance, effect sizes
- Evaluate model robustness under distribution shift, noise, or adversarial perturbation
- Report compute and efficiency honestly (parameters, FLOPs, latency, memory)
- Position your results against published work in your problem area
- Articulate limitations and failure modes without defensiveness

4. What You Will Submit

Four artifacts, all linked from your repository:

Deliverable	Description	Format
Updated Repository	All evaluation scripts under <code>src/evaluation/</code> , ablation configs, error analysis notebooks.	GitHub URL + tag
Final Results Table	A clean CSV with all final metrics: per-seed, per-class, mean and standard deviation.	CSV + report table
Error Analysis Notebook	Notebook documenting failure modes, hard examples, subgroup performance, and per-class diagnostics.	.ipynb + HTML export
Evaluation Report	Comprehensive written report on all evaluation activities.	PDF, 14–18 pages

5. Detailed Task Breakdown

Task 1: Final Test-Set Evaluation

Following the ceremony above, run the test evaluation. The deliverable for this task is a results table containing:

- Every metric you defined in your Phase 1 roadmap, evaluated on the test set
- Per-seed values (at least three seeds), plus mean and standard deviation
- Per-class breakdown for classification tasks (precision, recall, F1 per class)
- Comparison row: baseline (Phase 2 MLP) results on the same test set, also with three seeds
- Comparison row: deep model from Phase 3 on the same test set
- Comparison row: refined model from Phase 4 on the same test set

This table is the centrepiece of your report. It must be reproducible from a single command, documented in the README.

Task 2: Ablation Studies

An ablation is a controlled experiment that removes or replaces one component of your system to measure its contribution. You will run at least five ablations, chosen from the categories below. Each ablation reuses the same train/val split, the same seed strategy, and reports the same primary metric.

Ablation Type	Examples
Remove a component	Train without augmentation; without dropout; without batch normalization; without transfer learning (random init); without the generative-model contribution.
Replace a component	Swap Adam for SGD; replace LSTM with GRU; replace ResNet with VGG; swap GELU for ReLU.
Vary capacity	Halve the number of hidden units, double the depth, change the number of attention heads. Measure the capacity–performance curve.
Data ablation	Train on 10%, 25%, 50%, 100% of the training set. Produces a learning curve that tells you whether more data would help.
Training-recipe ablation	Without learning-rate scheduling; without warmup; without gradient clipping. Tests whether your training tricks earn their keep.

Required:

- At least five ablations spanning at least three of the categories above
- All ablations reported in a single table: configuration name, primary metric (mean $\pm$ std over seeds), parameter count, training time, delta vs full model

- Written interpretation for each ablation: did it surprise you? What does it imply about your model?
- Be open to negative findings: if removing a component does not hurt performance, that component does not earn its place. Report this; do not hide it.

Task 3: Error Analysis

Average metrics hide a lot. The point of error analysis is to look behind the mean and ask where the model fails, how, and why.

3a. Quantitative Error Analysis

- Confusion matrix (for classification) or residual plots (for regression). Normalized and unnormalized versions.
- Per-class metrics: precision, recall, F1, support. Identify the worst class and the best class.
- Calibration analysis: plot predicted probability vs actual accuracy in bins. A reliable model is well-calibrated.
- If your dataset has subgroups (gender, age bracket, language, geography, device, time period), compute metrics per subgroup. Disparities here are real findings, not afterthoughts.

3b. Qualitative Error Analysis

Pick 30–50 wrong predictions and look at them. By hand. There is no substitute for this.

- Cluster the errors into 4–6 categories: e.g., “model confused by occlusion”, “near-duplicate classes”, “noisy labels”, “out-of-distribution input”
- Estimate the proportion of each category in your error population
- Show 2–3 examples per category with model prediction and ground truth
- For at least one category, propose a concrete fix (even if you do not implement it)

3c. Hard-Example Mining

Identify the 20 examples your model is least confident about (highest entropy in the prediction distribution, or largest residuals for regression). Inspect them. Are they genuinely hard, mislabelled, or out-of-distribution? Report your assessment.

Task 4: Robustness Analysis

Real models face inputs that are noisier, dirtier, or simply different from the curated test set. Pick at least two robustness probes appropriate to your modality:

Modality	Robustness Probes
Images	Gaussian noise at varying levels, brightness/contrast shift, JPEG compression, occlusion patches, rotation, FGSM adversarial examples (epsilon = 0.01–0.1).
Text	Synonym substitution, typo injection, casing flips, removal of stopwords, sentence reordering, paraphrasing.

Modality	Robustness Probes
Tabular	Feature noise, missing-value injection, swapped categorical values, drift simulation (shift in feature distribution).
Audio / Time series	Additive noise, time stretching, pitch shifting (audio), missing timestamps, jittered sampling.

For each probe: plot performance as a function of perturbation strength. A graceful degradation curve is more informative than a single number.

Task 5: Efficiency & Compute Reporting

A model that takes 4 seconds per prediction is useless for many applications. Report:

- Parameter count (total and trainable)
- Approximate FLOPs per forward pass (use ptflops, thop, or equivalent)
- Average inference latency on your hardware (single sample and batched, with warmup runs excluded)
- Peak GPU memory during inference
- Total training time across the project (sum of all final-model runs, plus the HPO study from Phase 4)
- Disk size of the final checkpoint

Pareto frontier: if your ablations include capacity variants (Task 2), plot accuracy vs parameter count. Where on the Pareto frontier does your final model sit? Is there a smaller model that would be “good enough” for deployment?

Task 6: Comparison with Published Work

Revisit your Phase 1 literature review. Compare your final results to the published numbers.

- Find at least three published results on the same dataset (or a very similar one) and put them in a comparison table with yours
- Note when comparisons are not apples-to-apples: different splits, different metrics, different evaluation protocols, different compute scales
- Be honest about gaps. If a published method beats you by 8 points, explain why (more compute, more data, more sophisticated technique, different problem framing)
- If you outperform a published baseline, double-check the comparison fairness before claiming victory

Task 7: Statistical Reporting

Numbers without uncertainty estimates are folklore. Required:

- All headline metrics reported with mean and standard deviation over at least three seeds
- Bootstrap confidence intervals for at least the primary metric (1,000 resamples is enough)
- A paired comparison (paired t-test or McNemar test for classification) between your final model and the strongest baseline. State the test, the test statistic, and the p-value.
- Effect size, not just significance: a 0.2-point gain that is statistically significant on a huge test set is still a 0.2-point gain

6. The Limitations Section

Every Phase 5 report contains a Limitations section. This is graded positively, like the negative-results section in Phase 4. The list below is not exhaustive; address the ones that apply to your project.

- Dataset size and representativeness: who or what is underrepresented?
- Label quality: did you find evidence of label noise during error analysis?
- Distribution shift: how would the model behave on data from a different time, place, or device?
- Compute budget: what would more compute have enabled?
- Fairness or bias concerns: are there subgroups where the model performs worse?
- Failure modes that would be unacceptable in deployment, even if rare
- Confidence calibration issues: does the model express uncertainty appropriately?
- Threats to validity: anything that could make your reported numbers optimistic

A model whose limitations you can list precisely is a model you understand. Vagueness here costs marks.

7. Evaluation Report Structure

The report (14–18 pages, PDF) must contain:

1. Cover Page & Repository Link

Team, course, GitHub URL, commit hash and tag, date.

2. Executive Summary of Results

Half a page. The final test metric with confidence interval, the single biggest finding from the ablations, and the single most important limitation.

3. Test-Set Evaluation Protocol

How you froze the model, when you ran the test evaluation, exactly what you did. This proves the ceremony was followed.

4. Final Results Table

The headline table comparing baseline, Phase 3 model, Phase 4 refined model, and final model. All with mean $\pm$ std.

5. Ablation Studies

Five-ablation table plus per-ablation written interpretation. About 3 pages.

6. Error Analysis

Quantitative breakdown, qualitative categorization with examples, hard-example mining. About 3 pages.

7. Robustness Analysis

Probes used, degradation curves, interpretation. About 2 pages.

8. Efficiency Report

Parameters, FLOPs, latency, memory, training time, Pareto plot. About 1 page.

9. Comparison with Literature

Table and discussion. About 1 page.

10. Statistical Reporting

Confidence intervals, paired tests, effect sizes. Can be integrated into earlier sections or kept separate.

11. Limitations

Substantial. At least one full page of honest discussion.

12. Plan for Phase 6

What does the final week (deployment, presentation, final integration) need to address?

8. Submission Guidelines

- Tag your submission commit: `git tag phase5-submission && git push --tags`
- The final-test-evaluation tag from Task 1 must be visible in git history. We check that it predates phase5-submission and that the test evaluation script was not modified after it.
- Submit on the LMS: (a) PDF of evaluation report, (b) GitHub URL, (c) phase5-submission commit hash, (d) final-test-evaluation commit hash, (e) link to error analysis notebook
- Late penalty: 10% per day, no submissions after 72 hours late

9. Evaluation Rubric

Criterion	Weight
Test-set ceremony (procedure, single evaluation, documented)	10%
Ablation studies (rigour, interpretation, breadth)	20%
Error analysis (quantitative + qualitative)	20%
Robustness analysis	10%

Criterion	Weight
Efficiency reporting & Pareto thinking	5%
Literature comparison (fairness, honesty)	5%
Statistical reporting (CIs, tests, effect sizes)	10%
Limitations section (depth and honesty)	10%
Report quality (writing, figures, reproducibility)	10%

Notice that **zero percent is awarded for the absolute test-set number**. The grade is entirely about the rigour and honesty of the evaluation. A team reporting a modest result with deep analysis scores higher than a team reporting a strong number with shallow analysis. This is intentional and matches how the research community grades papers.

10. Common Pitfalls

- **Re-running the test set after a “small fix.”** Doesn’t matter how small the fix is. Once the test set is run, it is closed. Re-running invalidates the ceremony.
- **Ablating only “safe” components.** If every ablation confirms that everything matters, you probably weren’t ablating hard enough. Surprises are a sign of rigour, not failure.
- **Average metrics only.** Per-class, per-subgroup, per-seed breakdowns are where the real findings live.
- **Error analysis from 5 examples.** Five is a number from a tutorial. Thirty to fifty is a number from real analysis.
- **Adversarial examples without a baseline.** Adding noise and reporting the new accuracy means little if you don’t compare to (a) the clean accuracy and (b) how a simpler model degrades under the same noise.
- **Citing literature numbers as if they’re directly comparable.** Different splits, different metrics, different protocols. State the asymmetries explicitly.
- **“We ran out of time for ablations.”** Five small ablations on a held-out subset of the training data are tractable in two days. Plan compute around it.
- **Defensive limitations.** “Our model has limitations but they are minor” is not a limitations section. Real limitations are uncomfortable to write down. Write them anyway.